

Mathematical Proof IV: Integrated Proof Workshop¹

1 THE GOAL

In the first three sessions, we practiced three separate stages of mathematical problem solving:

Deconstruct	→	Explore	→	Construct
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Today we put them together.

The goal is to practice the entire process:

Read	→	Explore	→	Write	→	Check
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For each problem, we will ask:

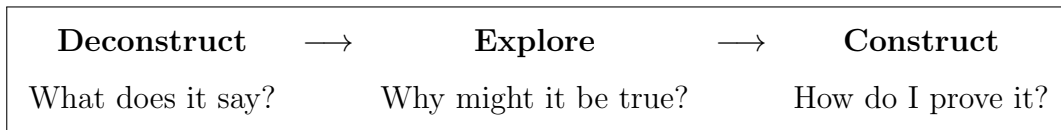
1. What exactly is given?
2. What exactly must be shown?
3. What useful consequences follow from the assumptions?
4. What would be sufficient to establish the goal?
5. What intermediate statement or construction could connect the two?
6. How should the proof be organized?
7. Is every step valid?

The amount of guidance will decrease throughout the session.

¹Instructors: Yue Yu and Sofia Olguin. This note was prepared for the 2026 UCSB Math Camp for Ph.D. students in economics. It incorporates materials from previous instructors, including Shu-Chen Tsao, ChienHsun Lin, and Sarah Robinson.

2 THE FULL WORKFLOW

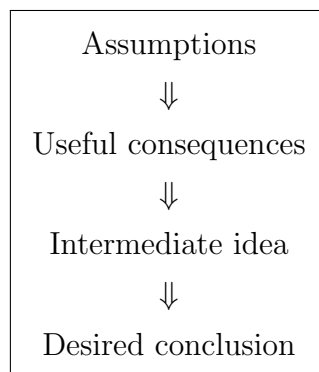
A useful mental model is:



followed by

Critique: Is the argument actually valid?

More explicitly:



The final step is to read the proof again and ask:

For every important line: Why does this follow?

3 WORKSHOP 1: GUIDED FULL PROCESS

We begin with a problem that resembles arguments you will see in consumer theory.

Proposition 1

Let

$$B(p, w) = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\},$$

where $p \gg 0$ and $w > 0$.

Suppose $u : B(p, w) \rightarrow \mathbb{R}$ is strictly increasing in every component and

$$x^* \in \operatorname{argmax}_{x \in B(p, w)} u(x).$$

Show that

$$p \cdot x^* = w.$$

3.1 Step 1: Deconstruct

What is given?

What exactly must be shown?

What does

$$x^* \in \operatorname{argmax}_{x \in B(p, w)} u(x)$$

tell us?

What does strict monotonicity tell us?

3.2 Step 2: Explore

Suppose the conclusion is false.

Then

$$p \cdot x^* < w.$$

Define the unused wealth:

$$\varepsilon = w - p \cdot x^* > 0.$$

What could we do with this slack?

Choose some good i , and let e_i denote the i -th standard basis vector.

Consider

$$x' = x^* + \delta e_i.$$

What condition on $\delta > 0$ would ensure that x' is still feasible?

What would strict monotonicity imply about $u(x')$ and $u(x^*)$?

Where would the contradiction occur?

3.3 Step 3: Construct

Write a proof skeleton first.

$$p \cdot x^* < w \implies \underline{\hspace{10cm}}$$

$$\implies \underline{\hspace{10cm}}$$

$$\implies u(x') > u(x^*)$$

$$\implies \text{contradiction to optimality.}$$

Now write the complete proof.

3.4 Step 4: Critique

Before moving on, check your argument.

- (a) Where did you use $p_i > 0$?
- (b) Where did you use strict monotonicity?
- (c) Where did you use optimality of x^* ?
- (d) Did you explicitly show that x' is feasible?

4 WORKSHOP 2: LESS GUIDANCE

Now consider a result from concave optimization.

Proposition 1

Let $C \subseteq \mathbb{R}^n$ be convex and let $f : C \rightarrow \mathbb{R}$ be concave.

Suppose

$$x^*, y^* \in \operatorname{argmax}_{x \in C} f(x).$$

Show that for every $\lambda \in [0, 1]$,

$$\lambda x^* + (1 - \lambda)y^*$$

is also a maximizer.

4.1 Part A: Deconstruct

Write:

Given:

Goal:

4.2 Part B: Explore

What follows from the fact that x^* and y^* are both maximizers?

What does convexity of C give you?

What does concavity of f give you?

What lower bound can you obtain for

$$f(\lambda x^* + (1 - \lambda)y^*)?$$

What upper bound follows from the fact that x^* and y^* are maximizers?

4.3 Part C: Construct

Write a proof plan using no more than four lines.

Now write the complete proof.

4.4 Part D: Critique

Check:

- (a) Did you show that the convex combination is feasible?
- (b) Did you obtain both a lower bound and an upper bound?
- (c) Did you explicitly conclude that the value equals the maximum?

5 WORKSHOP 3: MINIMAL SCAFFOLDING

Now solve the following almost entirely from scratch.

Proposition 1

Let $u : C \rightarrow \mathbb{R}$ be concave, where $C \subseteq \mathbb{R}^n$ is convex.

Show that u is quasiconcave.

You may use the definition:

u is quasiconcave $\iff u(\lambda x + (1 - \lambda)y) \geq \min\{u(x), u(y)\}$
for all $x, y \in C$ and $\lambda \in [0, 1]$.

Before writing the proof

Write a short plan.

What does concavity give you?

What intermediate inequality would connect concavity to quasiconcavity?

Write the proof.

Now check the proof.

Which step is the key bridge between the assumption and the goal?

6 WORKSHOP 4: PROVE OR DISPROVE

For the next problems, you are not told whether the statement is true.

For each one:

1. deconstruct the claim;
2. test simple examples;
3. decide whether to prove or disprove;
4. if true, write a proof;
5. if false, give a counterexample and explain exactly why the claim fails.

6.1 Claim 1

If f and g are increasing functions, then

$$h(x) = f(x)g(x)$$

is increasing.

6.2 Claim 2

If f and g are convex, then

$$f + g$$

is convex.

6.3 Claim 3

If A is positive semidefinite and

$$x^\top Ax = 0,$$

then

$$x = 0.$$

7 WORKSHOP 5: REPAIR THE STATEMENT

Sometimes a mathematical statement is almost correct, but one assumption is missing.

Consider:

Proposition 1

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable and

$$f'(x^*) = 0.$$

Then x^* is a global maximizer of f .

1. Is the proposition true?
2. Find a counterexample if it is false.
3. What additional assumption would make the statement true?
4. Write a corrected theorem.
5. Prove the corrected theorem.

8 WORKSHOP 6: A FIRST-YEAR MICRO STYLE PROBLEM

Consider the consumer problem

$$\max_{x \in B(p, w)} u(x),$$

where

$$B(p, w) = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}.$$

Assume:

$$p \gg 0, \quad w > 0,$$

the function u is continuous and strictly concave, and the budget set is nonempty.

Consider the claim:

Proposition 1

The consumer's optimization problem has a unique solution.

Your task is to build the argument from scratch.

8.1 Deconstruct

What two logically separate statements are contained in the phrase “has a unique solution”?

Hint:

Existence and Uniqueness.

8.2 Explore Existence

What theorem might establish existence of a maximizer?

What properties must the feasible set satisfy?

Is the budget set bounded when $p \gg 0$ and $w > 0$?

Explain.

Is the budget set closed?

Explain.

8.3 Explore Uniqueness

Suppose there were two distinct maximizers.

What does convexity of the budget set allow you to construct?

What does strict concavity imply?

Where does the contradiction occur?

8.4 Construct

Write the proof in two clearly separated parts:

Existence

Uniqueness

8.5 Critique

Check your proof carefully.

- (a) Which assumption guarantees existence?
- (b) Which assumption guarantees uniqueness?
- (c) Did strict concavity alone prove existence?

(d) Where did $p \gg 0$ matter?

(e) Did you prove both parts of the claim?

9 FINAL CHALLENGE: NO SCAFFOLDING

For the final problem, use the entire proof-solving process without prompts.

Proposition 1

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable and concave.

Show that if

$$\nabla f(x^*) = 0,$$

then x^* is a global maximizer of f .

You may use the first-order characterization of concavity:

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in \mathbb{R}^n.$$

Your work

10 SELF-ASSESSMENT

After finishing a proof, ask yourself:

1. Did I understand the claim correctly?
2. Did I identify what each assumption gives me?
3. Did I explore before writing?
4. Did I find a useful intermediate statement or construction?
5. Did I justify every important transition?
6. Did I use every important assumption appropriately?
7. Did I prove exactly what was asked?
8. Can I explain the proof in one or two sentences?

11 THE COMPLETE PROOF-SOLVING ROUTINE

The four sessions can now be summarized as one routine.

1. Deconstruct

What is given? What must be shown?

**2. Explore**

What follows? What would be enough?

Try examples, counterexamples, and intermediate steps.

**3. Construct**

Write a proof skeleton and fill in the logical connections.

**4. Critique**

Why does each line follow? Did I prove the whole claim?

12 TAKEAWAY

The long-run goal is not to mechanically run through a checklist.

The goal is for this sequence of questions to become automatic:

What does the problem say?

What can I do with the assumptions?

What would be enough to reach the goal?

How do I turn that idea into a proof?

Is every step actually valid?

Eventually, solving a proof problem should feel less like searching for a trick and more like systematically building an argument.